Practical 3

Field	Details
Name of Student	Danish Shaikh
Roll No. of Student	329
Semester	B.Sc (I.T) SEM-V
Academic Year	2026-2027
Subject	**Data Visualization with Power BI and Tableau**
Faculty In-charge	Mr. Shahid Ansari
Practical No.	3
Title	**Practical 3: Creating Advanced Visualizations in Tableau** a) Create stacked and grouped bar charts for comparison b) Build tree maps and pie charts for part-to-whole analysis c) Design scatter plots to identify relationships d) Create dual-axis and combination charts e) Develop histograms for distribution analysis f) Build box-and-whisker plots g) Use Gantt charts and advanced date visualizations
Date	

Creating Advanced Visualizations in Tableau
a) Create stacked and grouped bar charts for comparison

a) Create Stacked and Grouped Bar Charts for Comparison

These two charts are useful for comparing **multiple categories and sub-categories**.

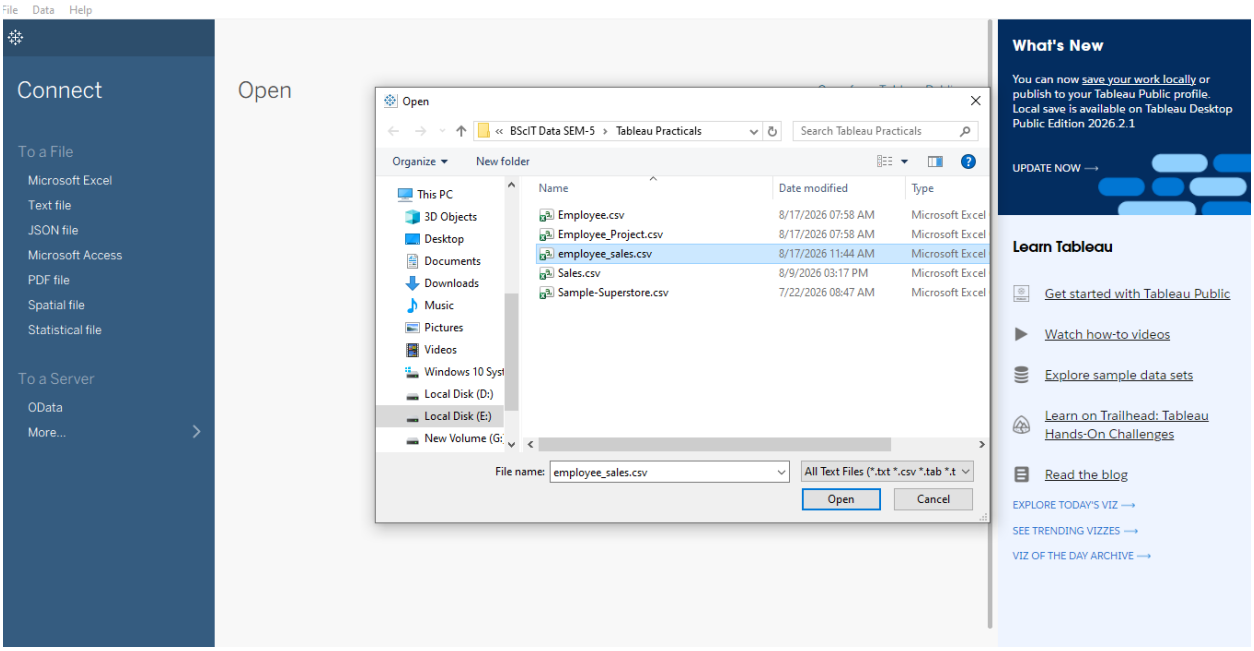
1. Stacked Bar Chart

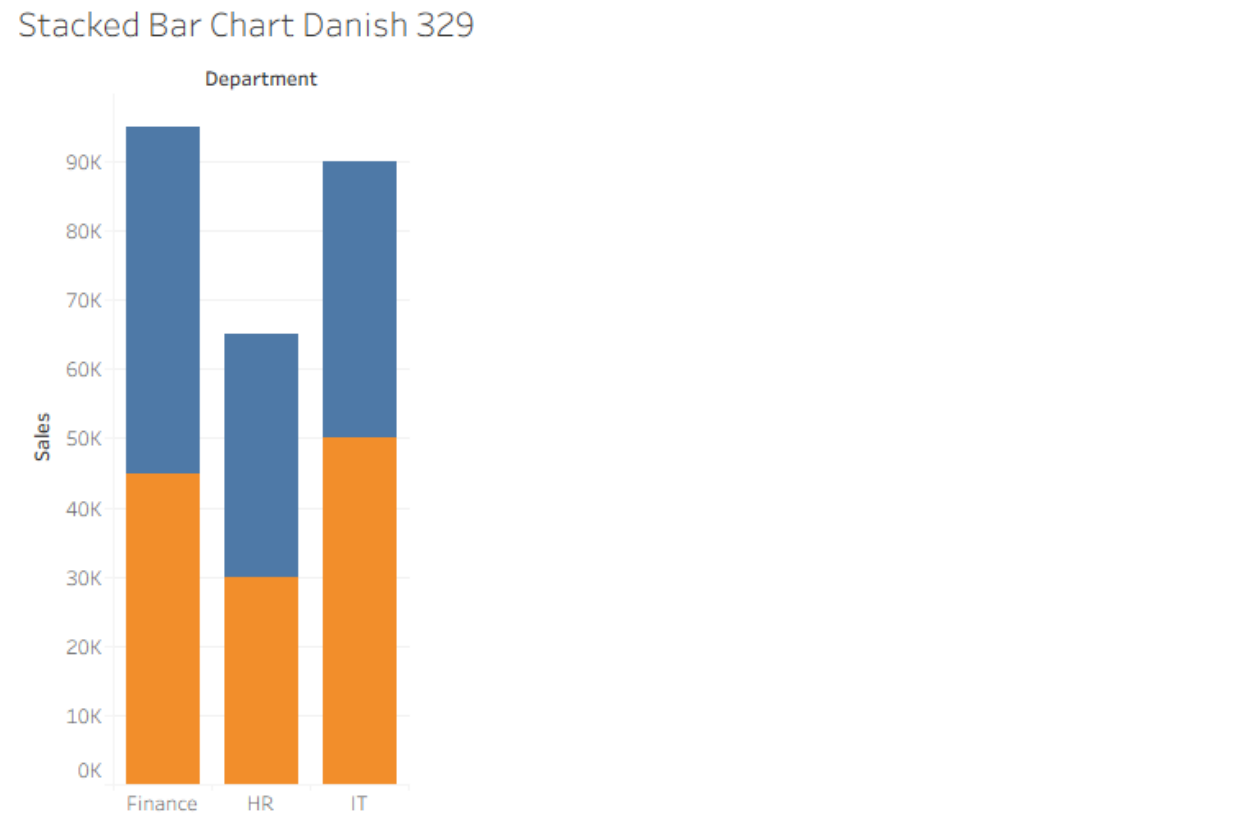
A **stacked bar chart** divides each bar into colored sections, allowing you to see both the **total value** and the contribution of each category.

Steps in Tableau

1. Open your Tableau workbook.
2. Go to a new **Worksheet**.
3. Drag **Department → Columns**.
4. Drag **Sales → Rows**.
5. Drag **Gender → Color** in the **Marks** card.
6. Make sure the **Marks** type is **Bar**.

You will get a chart where each department's bar is divided into **Male** and **Female** sections.





Grouped Bar Chart

A **grouped bar chart** places bars **side by side** instead of stacking them.

It is useful when you want to directly compare individual values.

Steps in Tableau

Using the same data:

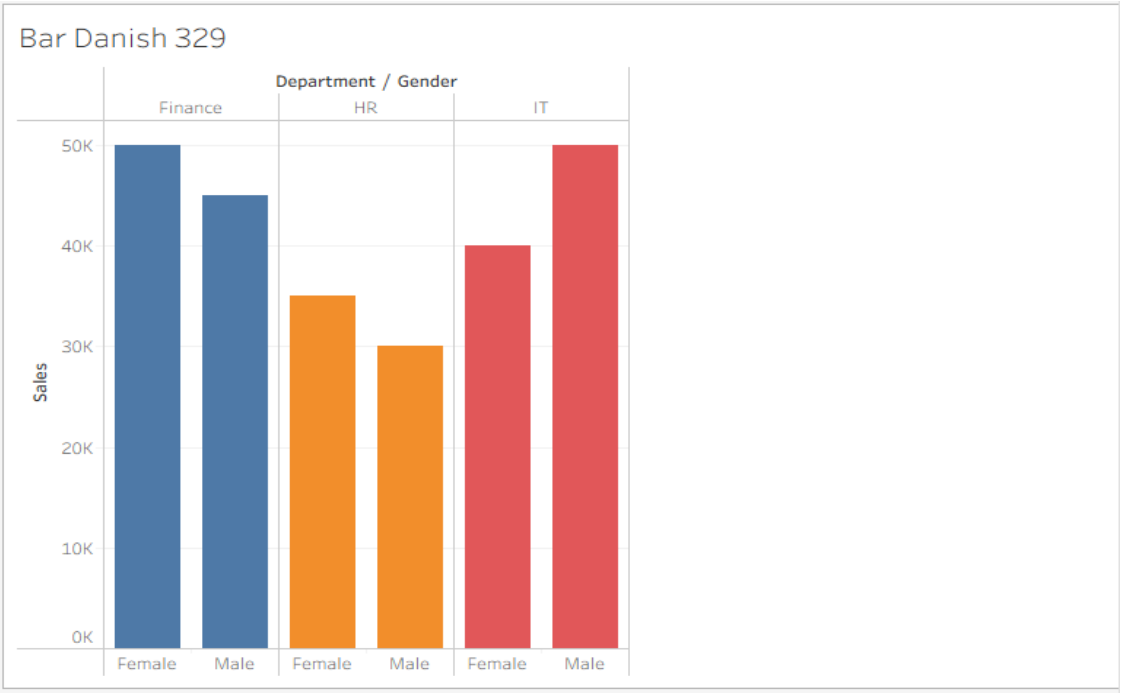
1. Drag **Department → Columns**.
2. Drag **Sales → Rows**.
3. Drag **Gender → Columns**, next to Department.
4. Keep **Marks** type as **Bar**.

The result will show:

- IT → Male | Female
- HR → Male | Female
- Finance → Male | Female

Stacked:
Department → Columns
Sales → Rows
Gender → Color

Grouped:
Department → Columns
Gender → Columns
Sales → Rows



Create a Treemap in Tableau

A treemap is useful for **part-to-whole analysis** because the size of each rectangle represents the value.

Steps

1. Open a **new Worksheet**.
2. In the **Data** pane, select:
    - Department
    - Sales
3. Click **Show Me** in the top-right.
4. Select **Treemap**.

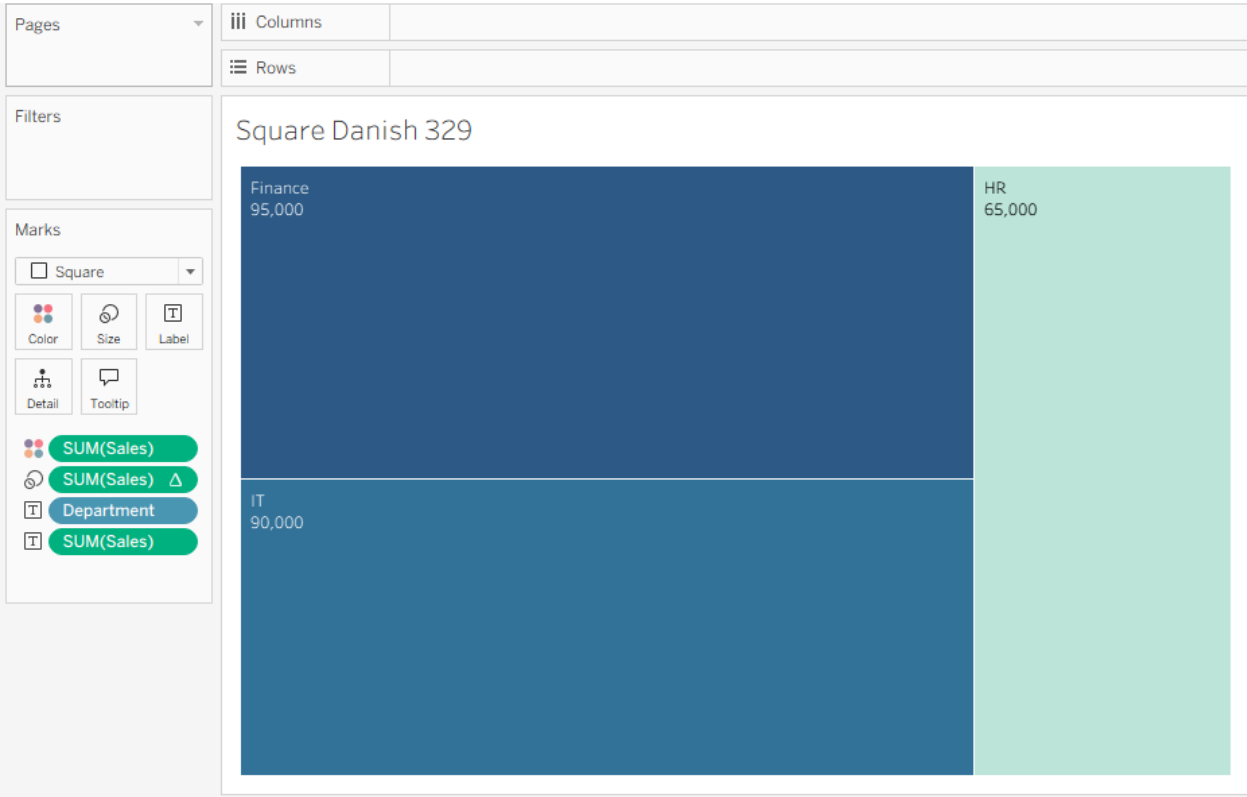
Alternatively, manually:

- Drag **Department → Marks → Label**
- Drag **Sales → Marks → Size**
- Drag **Sales → Marks → Color**
- Change **Marks** type to **Square**

You should get three rectangles:

- Finance = ₹95,000
- IT = ₹90,000
- HR = ₹65,000

The larger rectangle represents the department with higher total sales.



Create a Pie Chart

A **Pie Chart** in Tableau is used to show how individual categories contribute to a **whole**. It is especially useful when you want to answer questions such as:

- What percentage of total sales comes from each category?
- Which product category contributes the most?
- How is total profit distributed among regions?
- What is the market share of each product?

Steps in Tableau

1. Open Tableau

- **Open Tableau Desktop.**
- **Click Connect → Text File.**
- **Select your employee_sales.csv file.**
- **Click Sheet 1.**

2. Create the Pie Chart

1. **On the left, find Department.**
2. **Drag Department to Color in the Marks card.**
3. **Change the Marks type from Automatic to Pie.**
4. **Drag Sales to Angle.**
5. **Drag Sales again to Label.**

You should now see a pie chart showing sales distribution by department.

Your pie chart will show the contribution of:

Department	Sales	Approx. Share
Finance	95,000	36.5%
IT	90,000	34.6%
HR	65,000	25.0%

Total Sales = 250,000

Show Sales Percentage
To show each department's percentage of total sales:

1. Right-click the second **Sales** pill on **Label**.
2. Select **Quick Table Calculation**.
3. Choose **Percent of Total**.

Now the labels will show percentages such as:

- IT — 30%
- HR — 15%
- Sales — 40%
- Finance — 15%

Step 1: Select Pie

In your **Marks** card, click:

Automatic → Pie

Step 2: Add Department

Drag:

Department → Color

Step 3: Add Sales to Angle

Drag:

Sales → Angle

Your pie chart should now appear.

Step 4: Add Sales to Label

This is important.

Drag **Sales** from the left Data pane **again** and drop it on:

Label

You should now see a small pill like:

SUM(Sales)

inside the **Label** section.

Step 5: Open the Sales menu

Now **don't right-click the word "Label."**

Instead, find the small **SUM(Sales) pill** inside the Label area.

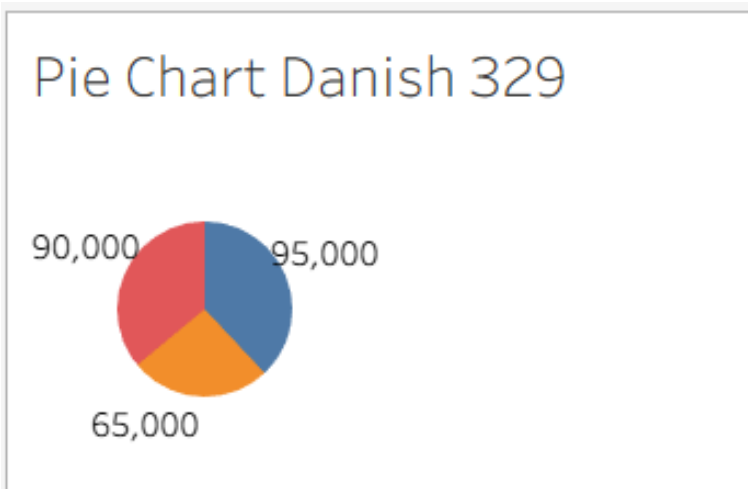
Click the **small down arrow** on that pill.

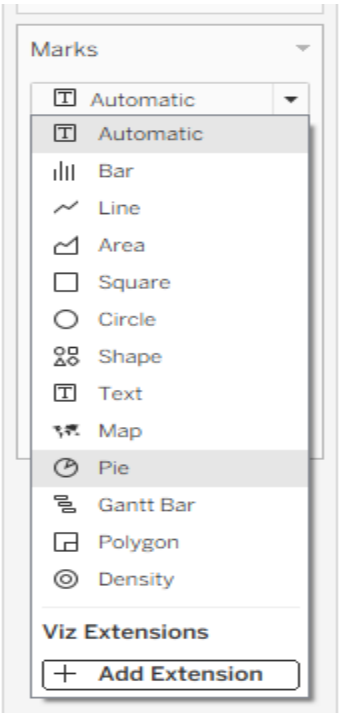
Quick Table Calculation → Percent of Total

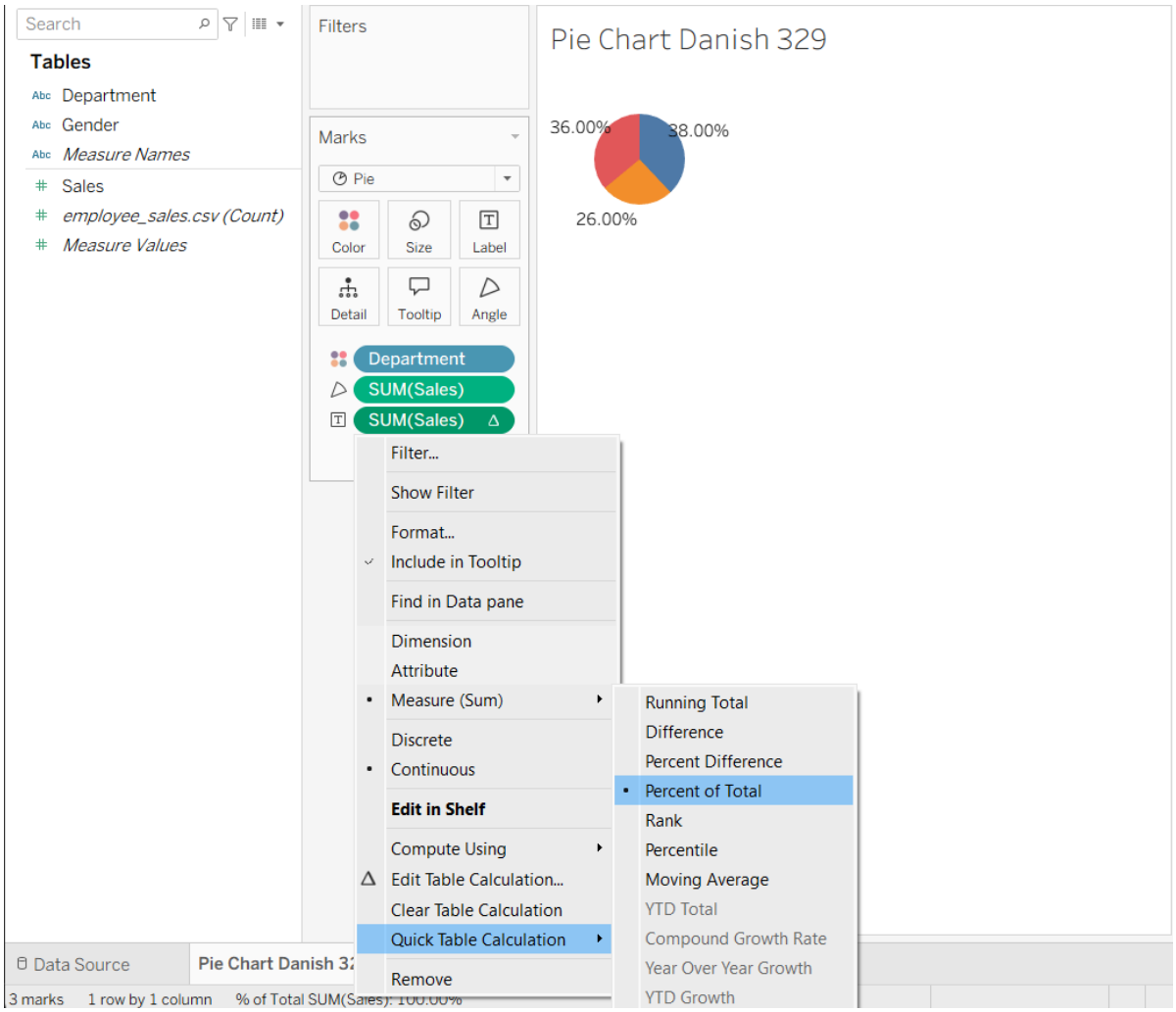
If you still don't see "Quick Table Calculation"

Try this alternative:

1. Click the small arrow on SUM(Sales) under **Label**.
2. Look for **Add Table Calculation...**
3. Select it.
4. For **Calculation Type**, choose:
   Percent of Total
5. Click **OK**.


Pie Chart Danish 329
90,000
95,000
65,000


Marks
Automatic
Automatic
Bar
Line
Area
Square
Circle
Shape
Text
Map
Pie
Gantt Bar
Polygon
Density
Viz Extensions
Add Extension


Search
Tables
Department
Gender
Measure Names
Sales
employee_sales.csv (Count)
Measure Values
Filters
Marks
Pie
Color
Size
Label
Detail
Tooltip
Angle
Department
SUM(Sales)
SUM(Sales)
Pie Chart Danish 329
36.00%
38.00%
26.00%
Filter...
Show Filter
Format...
Include in Tooltip
Find in Data pane
Dimension
Attribute
Measure (Sum)
Discrete
Continuous
Edit in Shelf
Compute Using
Edit Table Calculation...
Clear Table Calculation
Quick Table Calculation
Remove
Running Total
Difference
Percent Difference
Percent of Total
Rank
Percentile
Moving Average
YTD Total
Compound Growth Rate
Year Over Year Growth
YTD Growth
Data Source
Pie Chart Danish 32
3 marks
1 row by 1 column
% of Total SUM(Sales): 100.00%

Scatter Plot: Sales vs Number of Employees

Step 1: Create a new worksheet

At the bottom, click the + icon to create a new sheet.

Step 2: Put Sales on Columns

From the left Data pane:

Drag **Sales** to **Columns**

Put Count on Rows

From the left:

Drag **employee_sales.csv (Count...)** to **Rows**

Tableau will create a visualization.

Step 4: Change it to Scatter Plot

Look at **Show Me** on the top-right.

Click the **Scatter Plot icon**.

It looks like several individual dots:

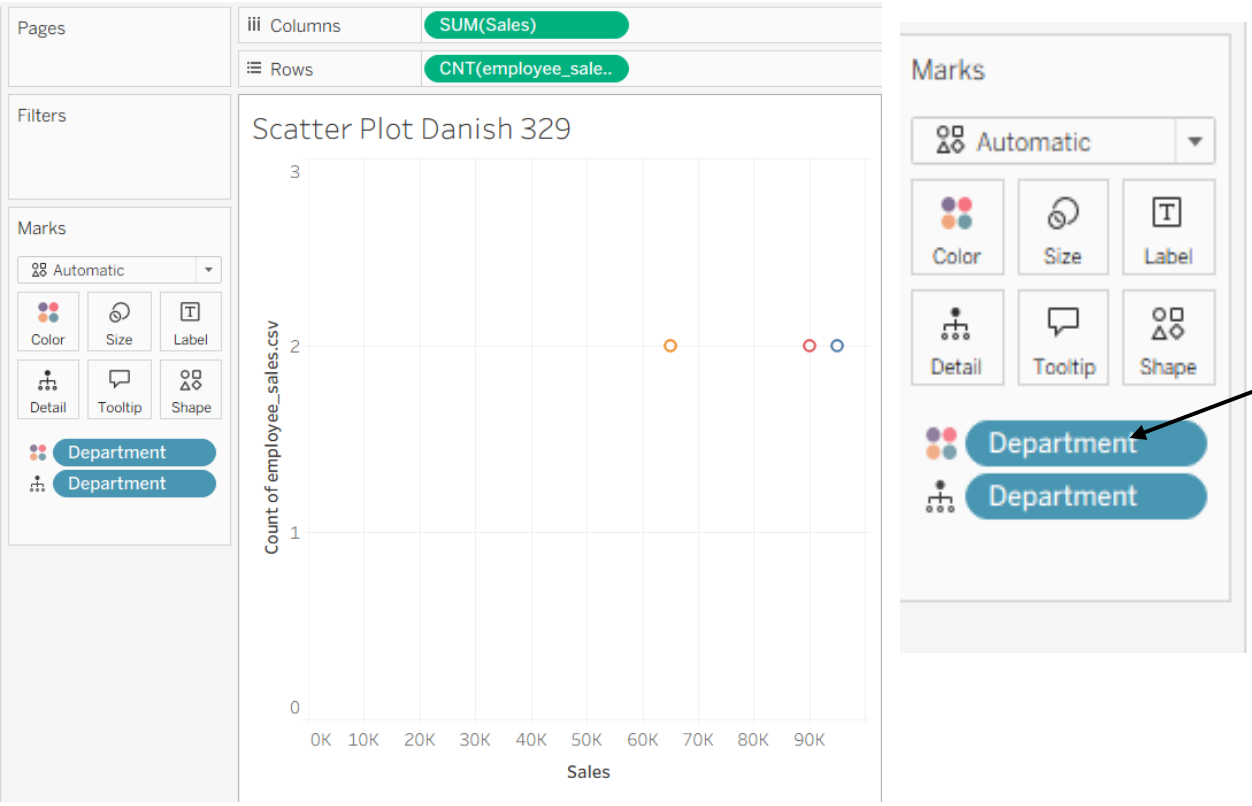
Do **not drag** the icon. Just **click it**.

Add Department to Detail

To make the points represent departments:

Drag **Department → Detail** in the Marks card.

You can also drag **Department → Color**.



Create a Dual-Axis Chart

We will compare **Sales** and **Advertising Cost**.

Step 1: Open Tableau

1. Open Tableau.
2. Connect your Excel file containing the dataset.
3. Go to a new worksheet.

Step 2: Add Month

1. Drag **Month** to the **Columns** shelf.

Step 3: Add Sales

1. Drag **Sales** to the **Rows** shelf.

A line chart for Sales will appear.

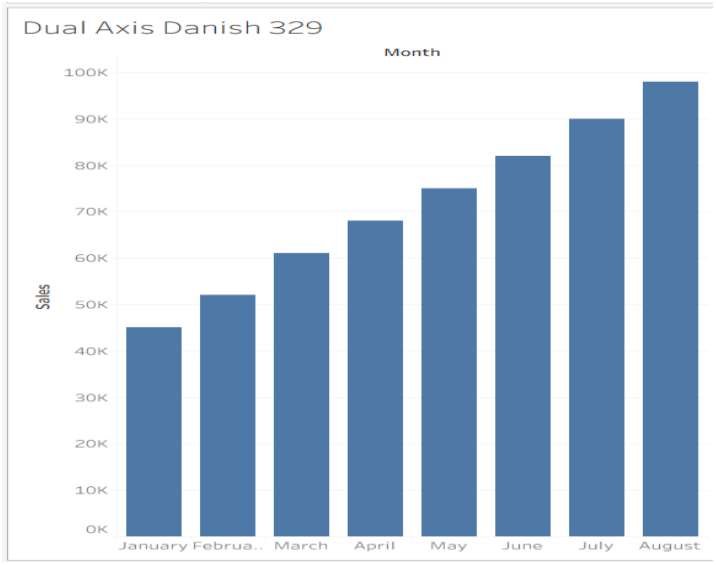
Step 4: Add Advertising Cost

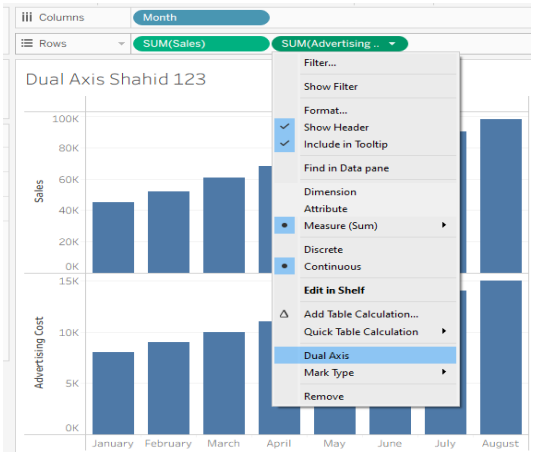
1. Drag **Advertising Cost** to the **Rows** shelf.
2. You will now have two charts.

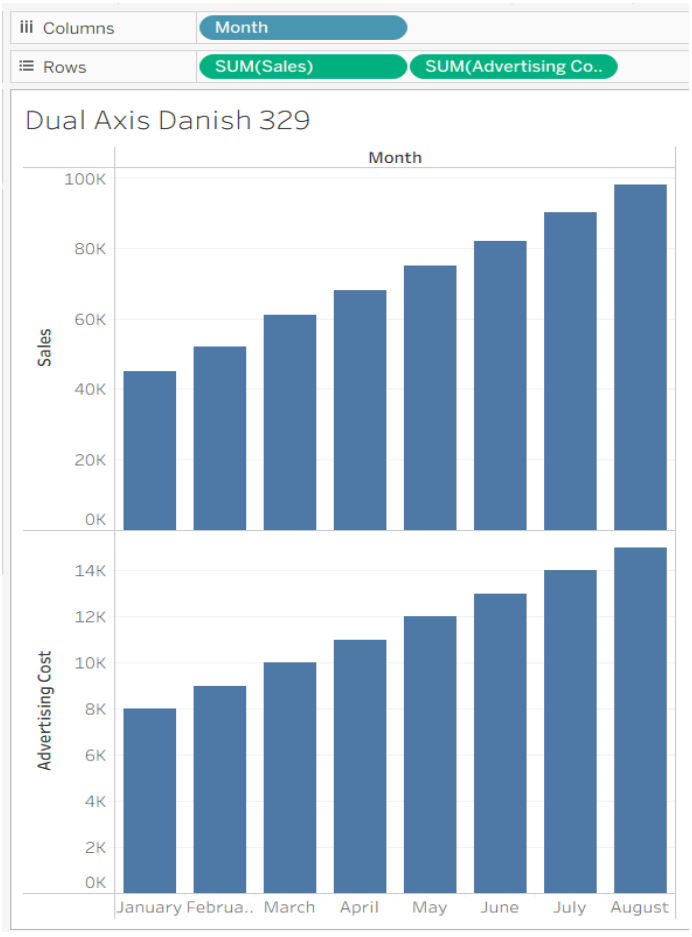
Step 5: Create Dual Axis

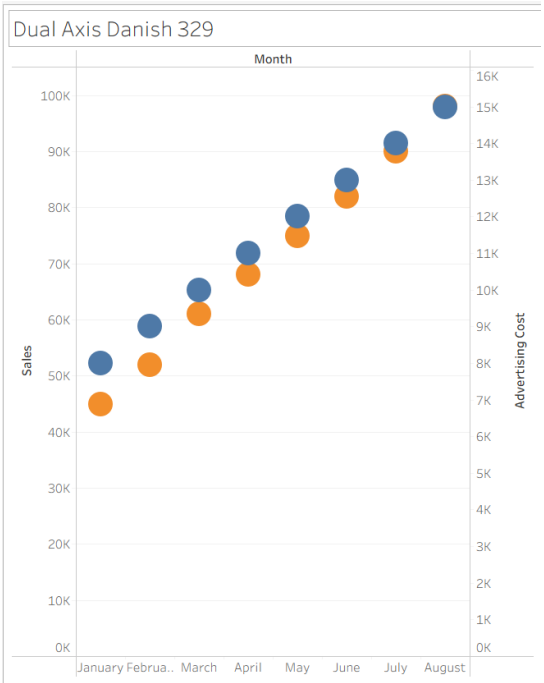
1. Right-click the second **Advertising Cost** axis.
2. Select **Dual Axis**.

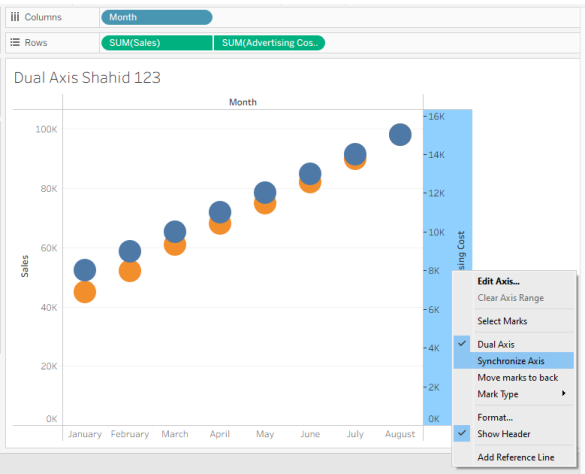
Both measures will now appear in the same view.

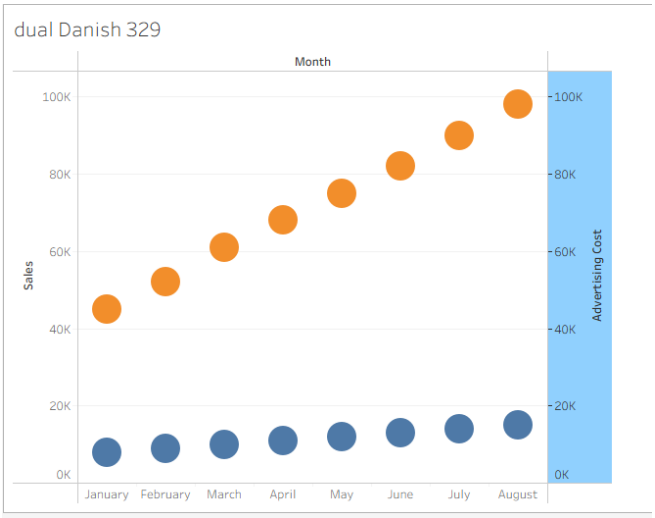












Create Another Combination Chart

compare **Visitors** and **Sales**.

Step 1

Drag **Month** to **Columns**.

Step 2

Drag **Visitors** to **Rows**.

Step 3

Drag **Sales** to the **Rows** shelf next to Visitors.

Step 4

Right-click the second measure's axis.

Select **Dual Axis**.

Step 5

Right-click an axis and select:

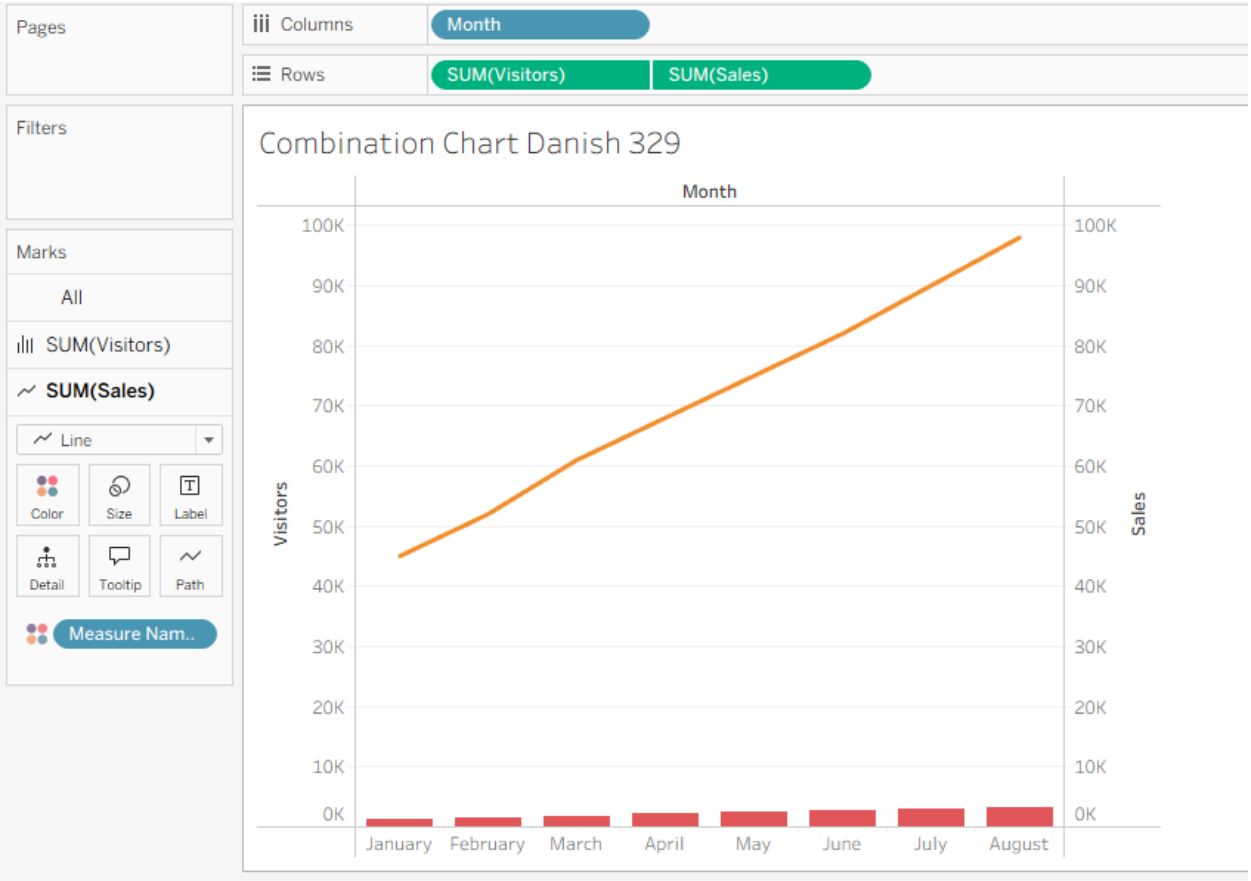
Synchronize Axis

Step 6

On the **Marks** cards:

- **Visitors → Bar**
- **Sales → Line**

This creates a combination chart showing website visitors as bars and sales as a line.



Develop Histograms for Distribution Analysis in Tableau

Create a Histogram of Visitors

Step 1: Open Tableau

1. Open Tableau.
2. Connect the CSV/Excel dataset.
3. Open a new worksheet.

Step 2: Select Visitors

1. Find **Visitors** in the Data pane.
2. Right-click **Visitors**.
3. Select **Create → Bins**.

Step 3: Set Bin Size

A **Create Bins** window will appear.

Set:

Size of bins = 500

Click **OK**.

Tableau will create a new field called something like:

Visitors (bin)

Step 4: Create the Histogram

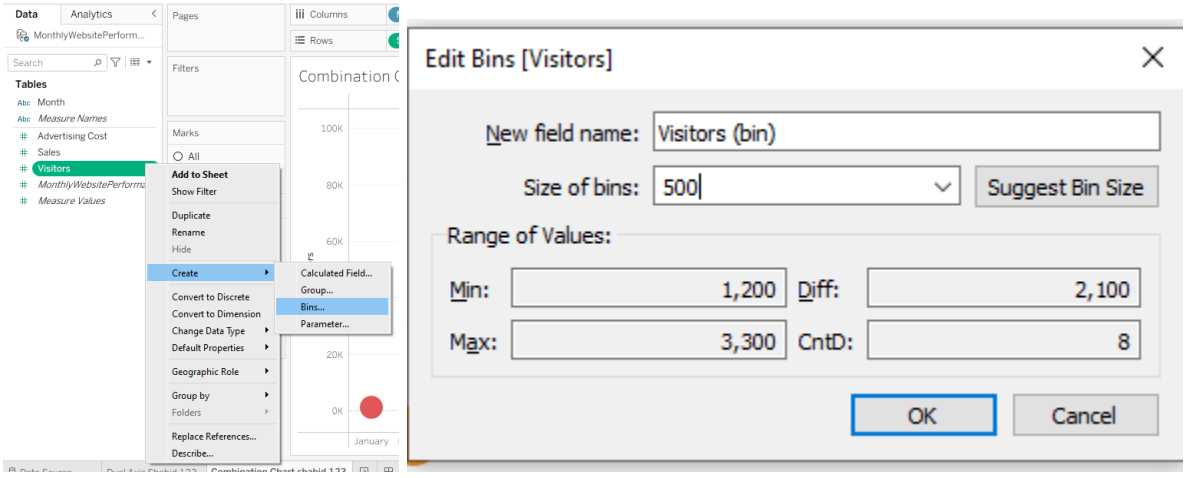
1. Drag **Visitors (bin)** to **Columns**.
2. Drag **Visitors** to **Rows**.
3. Tableau will display the frequency of visitors in each range.

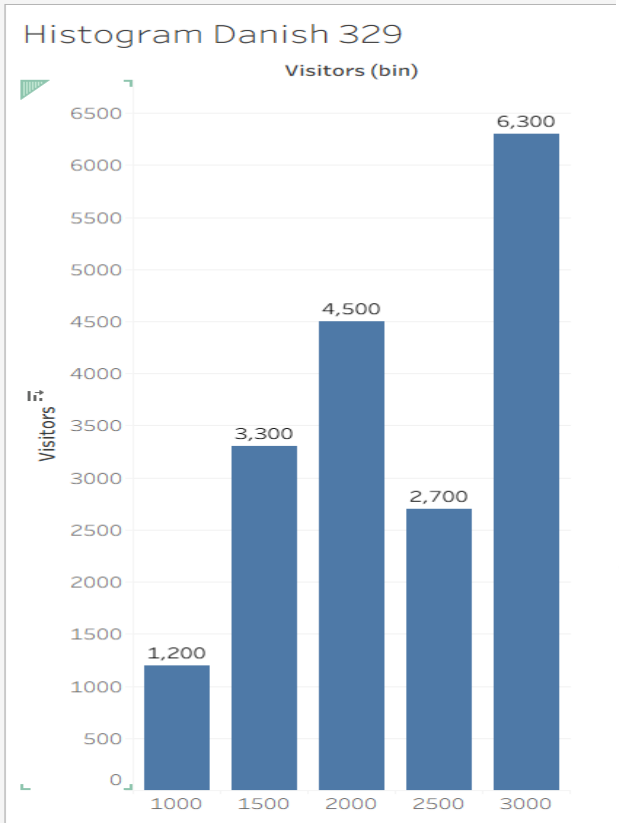
You now have a **Histogram**.

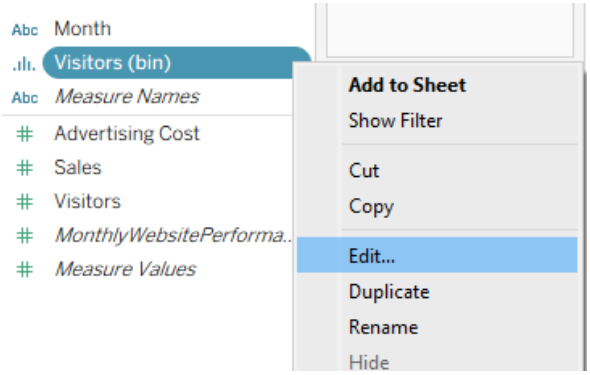
Step 5: Add a Title

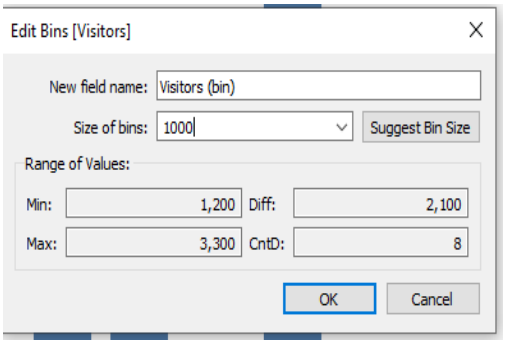
1. Go to **Worksheet → Show Title**.
2. Change the title to:

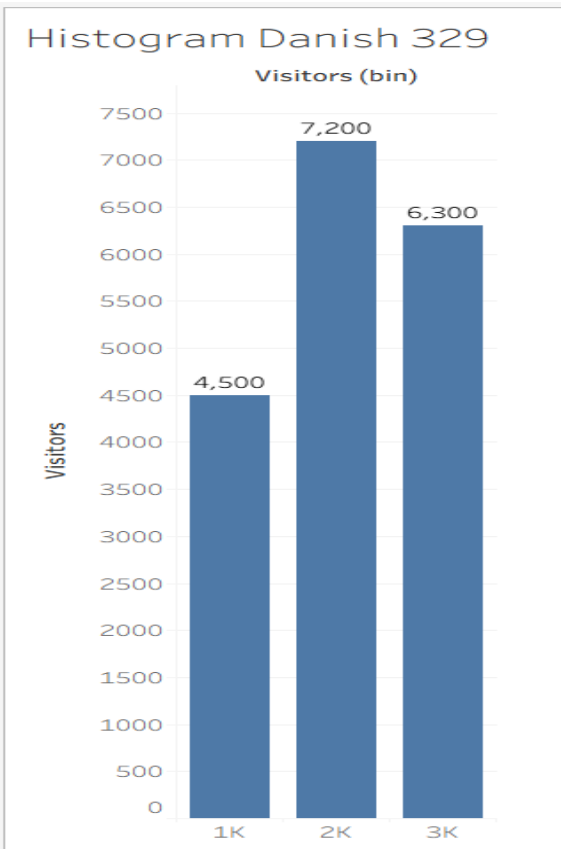
Distribution of Website Visitors




Histogram Danish 329
Visitors (bin)
Visitors
6500
6000
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
0
1,200
3,300
4,500
2,700
6,300
1000
1500
2000
2500
3000


Abc Month
Visitors (bin)
Abc Measure Names
Advertising Cost
Sales
Visitors
MonthlyWebsitePerforma..
Measure Values
Add to Sheet
Show Filter
Cut
Copy
Edit...
Duplicate
Rename
Hide


Edit Bins [Visitors]
New field name: Visitors (bin)
Size of bins: 1000
Suggest Bin Size
Range of Values:
Min: 1,200
Diff: 2,100
Max: 3,300
CntD: 8
OK
Cancel


Histogram Danish 329
Visitors (bin)
Visitors
7500
7000
6500
6000
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
0
4,500
7,200
6,300
1K
2K
3K

f) Box-and-Whisker Plot — Exact Steps

Step 1

Drag:

Sales → Rows

You should now see Visitors values for each month.

Step 2

On the Marks:

Month→ Details

Step 3

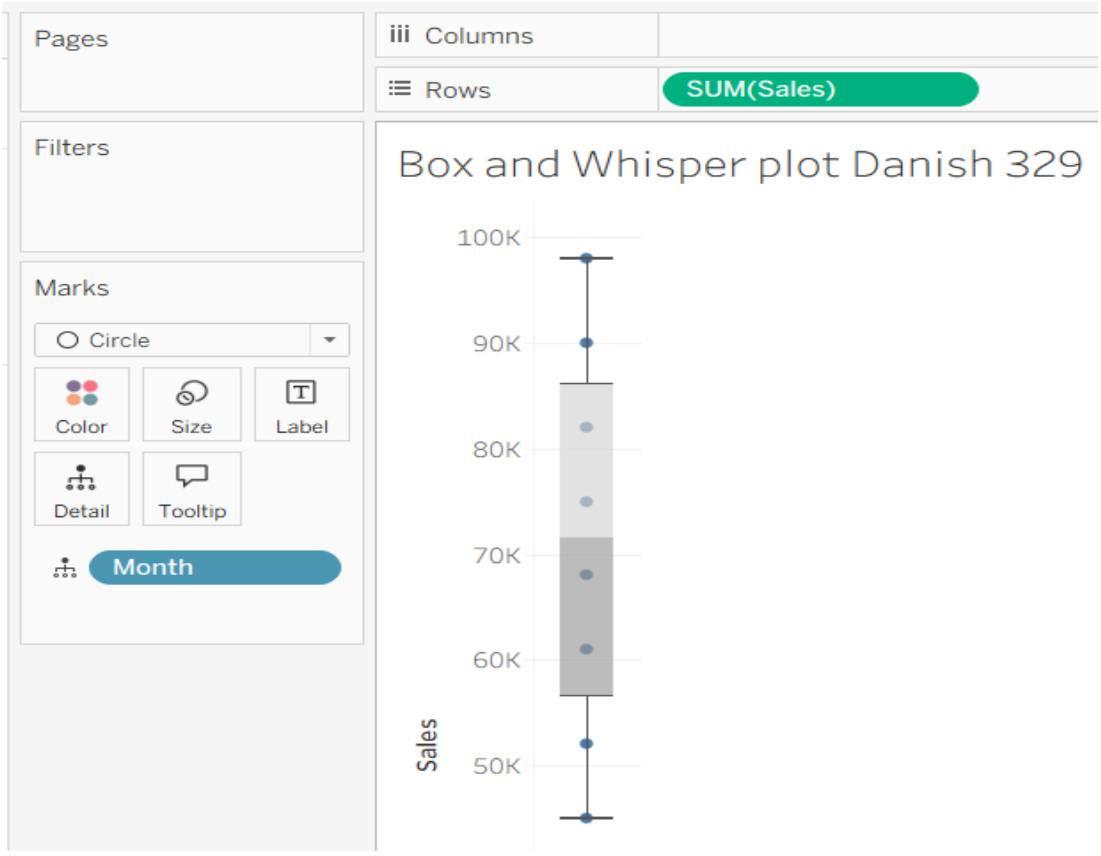
Click Show Me in the top-right.

Find and click the Box-and-Whisker Plot icon.

Then drag Box Plot onto Cell.

Step 4

Your chart should now show the distribution of Visitors by Month.



Steps to Create a Gantt Chart:

1. First, drag **SUM(Sales)** to the **Columns** shelf.

2. Then, drag **Month** to the **Rows** shelf.

3. Drag **Sales** to **Color**.

4. Drag **Sales** to **Detail**.

5. Drag **Sales** to **Size**.

6. From the **Marks** card, select **Gantt Bar**.

7.Change Column Sum Sales to Advertising Cost .



Practical 4

Field	Details
Name of Student	Danish Shaikh
Roll No. of Student	329
Semester	B.Sc (I.T) SEM-V
Academic Year	2026-2027
Subject	**Data Visualization with Power BI and Tableau**
Faculty In-charge	Mr. Shahid Ansari
Practical No.	4
Title	**Practical 4: Performing Data Analysis using Calculations in Tableau** a) Create basic calculated fields using formulas b) Perform row-level calculations c) Perform aggregate-level calculations d) Implement FIXED Level of Detail (LOD) calculations e) Use INCLUDE and EXCLUDE LOD expressions f) Create and use parameters for dynamic analysis g) Apply calculations to fix data issues and enhance insights
Date	

Aim

To perform basic data analysis in **Tableau** by creating calculated fields using formulas.

Dataset Example

You can use a simple **Sales** dataset with these columns:

Product	Category	Sales	Quantity	Profit
Laptop	Electronics	50000	2	8000
Mobile	Electronics	30000	3	6000
Chair	Furniture	10000	5	2000
Table	Furniture	20000	2	3000
Headphones	Electronics	5000	10	1500

Step 1: Open Tableau

1. Open **Tableau Desktop**.

2. Connect to your Excel/CSV dataset.
3. Open a **Worksheet**.

Step 2: Create a Calculated Field

In Tableau:

Analysis → Create Calculated Field

Give the calculated field a name and enter the formula.

Calculated Field Name: Total Amount

[Sales] * [Quantity]

This calculates the total amount based on sales and quantity.

Example 2: Profit Percentage

Calculated Field Name: Profit Percentage

([Profit] / [Sales]) * 100

This calculates profit as a percentage of sales.

Example 3:

Sales Category

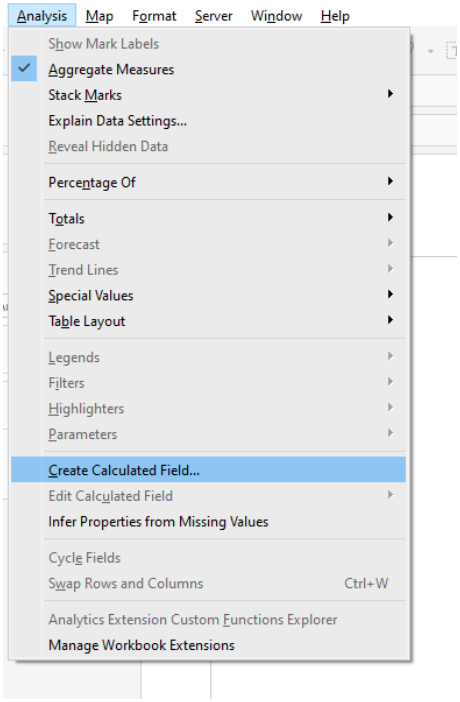
You can classify sales using an IF condition:

IF [Sales] >= 30000 THEN "High"

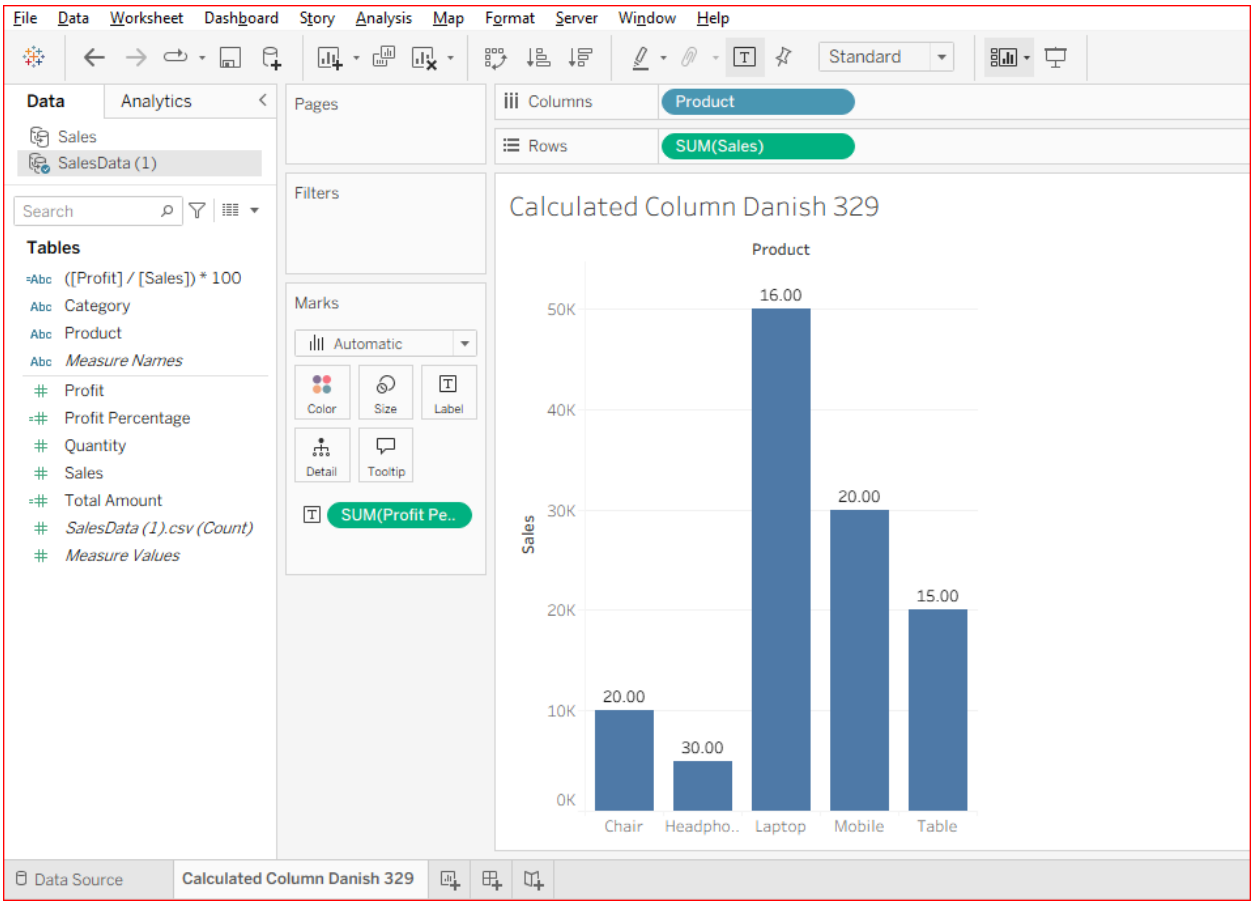
ELSEIF [Sales] >= 10000 THEN "Medium"

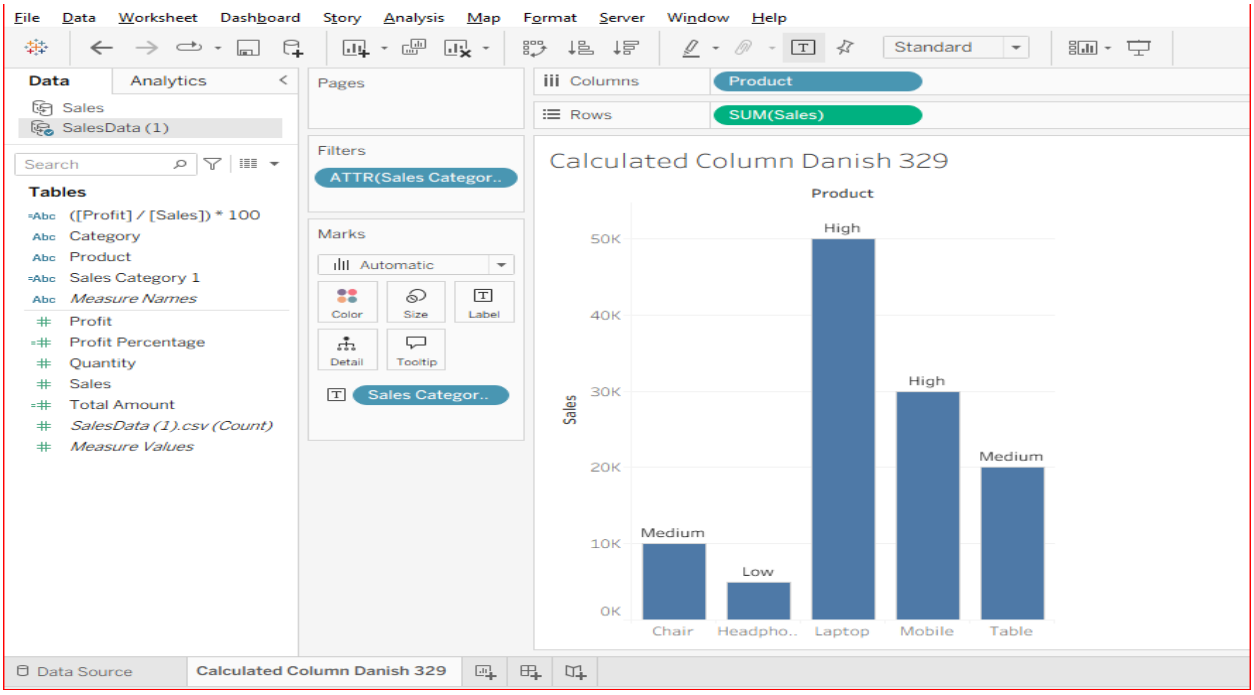
ELSE "Low"

END


Analysis
Map
Format
Server
Window
Help
Show Mark Labels
Aggregate Measures
Stack Marks
Explain Data Settings...
Reveal Hidden Data
Percentage Of
Totals
Forecast
Trend Lines
Special Values
Table Layout
Legends
Filters
Highlighters
Parameters
Create Calculated Field...
Edit Calculated Field
Infer Properties from Missing Values
Cycle Fields
Swap Rows and Columns
Ctrl+W
Analytics Extension Custom Functions Explorer
Manage Workbook Extensions

Output


File
Data
Worksheet
Dashboard
Story
Analysis
Map
Format
Server
Window
Help
Standard
Data
Analytics
Sales
SalesData (1)
Search
Tables
([Profit] / [Sales]) * 100
Category
Product
Measure Names
Profit
Profit Percentage
Quantity
Sales
Total Amount
SalesData (1).csv (Count)
Measure Values
Pages
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
SUM(Profit Pe..
Columns
Product
Rows
SUM(Sales)
Calculated Column Danish 329
Product
Sales
50K
40K
30K
20K
10K
0K
16.00
20.00
15.00
20.00
30.00
Chair
Headpho..
Laptop
Mobile
Table
Data Source
Calculated Column Danish 329



Perform Row-Level Calculations in Tableau

A **row-level calculation** performs a calculation **for each individual row of data** before Tableau performs aggregation.

Step 1: Create a Calculated Field

In Tableau:

Data pane → Right-click → Create → Calculated Field

Name it:

Sales per Quantity

Enter:

[Sales] / [Quantity]

Click **OK**.

Step 1: Create a Calculated Field

In Tableau:

Data pane → Right-click → Create → Calculated Field

Name it:

Sales per Quantity

Enter:

[Sales] / [Quantity]

Click **OK**.

Step 2: Understand the Calculation

Tableau calculates it **row by row**:

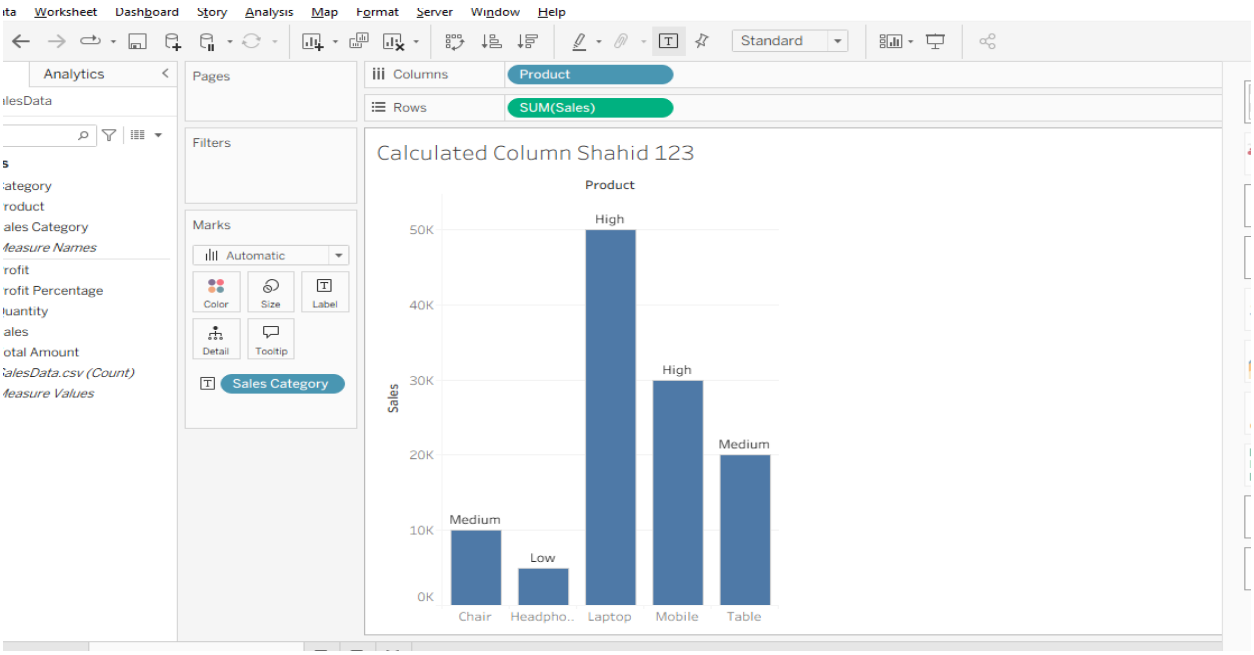
Product	Sales	Quantity	Sales per Quantity
Laptop	50000	2	25000
Mobile	30000	3	10000
Chair	10000	5	2000
Table	20000	2	10000

So, for example:

Laptop:

50000 / 2 = 25000

Thus, row-level calculations were successfully performed in Tableau using calculated fields, and the calculated values were used for data analysis.



Perform Aggregate-Level Calculations in Tableau

To perform **aggregate-level calculations** in Tableau using functions such as SUM(), AVG(), MIN(), and MAX().

What is an Aggregate-Level Calculation?

An aggregate calculation performs a calculation on a **group of rows** rather than on each individual row.

For example:

SUM([Sales])

calculates the **total sales** for the selected group.

Step 1: Create a Calculated Field

Go to:

Data Pane → Right-click → Create → Calculated Field

Enter:

Total Sales

SUM([Sales])

Click **OK**.

Step 2: Calculate Average Sales

Create another calculated field:

Average Sales

AVG([Sales])

Calculate Profit Percentage

You can calculate the overall profit percentage using:

SUM([Profit]) / SUM([Sales])

Format the result as **Percentage**.

For the above data:

Total Profit = 8000 + 6000 + 2000 + 3000 = 19000

Total Sales = 50000 + 30000 + 10000 + 20000 = 110000

Therefore:

Profit Percentage = 19000 / 110000

= 17.27%

Common Aggregate Functions

Function	Purpose	Example
SUM()	Total value	SUM([Sales])
AVG()	Average value	AVG([Sales])
MIN()	Minimum value	MIN([Sales])
MAX()	Maximum value	MAX([Sales])
COUNT()	Number of records	COUNT([Sales])
COUNTD()	Number of unique values	COUNTD([Product])

Implement FIXED Level of Detail (LOD) Calculations in Tableau

What is FIXED LOD?

A **FIXED LOD** tells Tableau to calculate a value based on a specific dimension, regardless of what dimensions are currently displayed in the visualization.

The basic syntax is:

{ FIXED [Dimension] : AGGREGATION([Measure]) }

Step 1: Create a FIXED LOD Calculated Field

Go to:

Data Pane → Right-click → Create → Calculated Field

Give it the name:

Category Total Sales

Enter:

{ FIXED [Category] : SUM([Sales]) }

Click **OK**.

Step 2: Understand the Result

Tableau calculates the total sales separately for each category.

Product	Category	Sales	Category Total Sales
Laptop	Electronics	50000	80000
Mobile	Electronics	30000	80000

Chair	Furniture	10000	30000
Table	Furniture	20000	30000

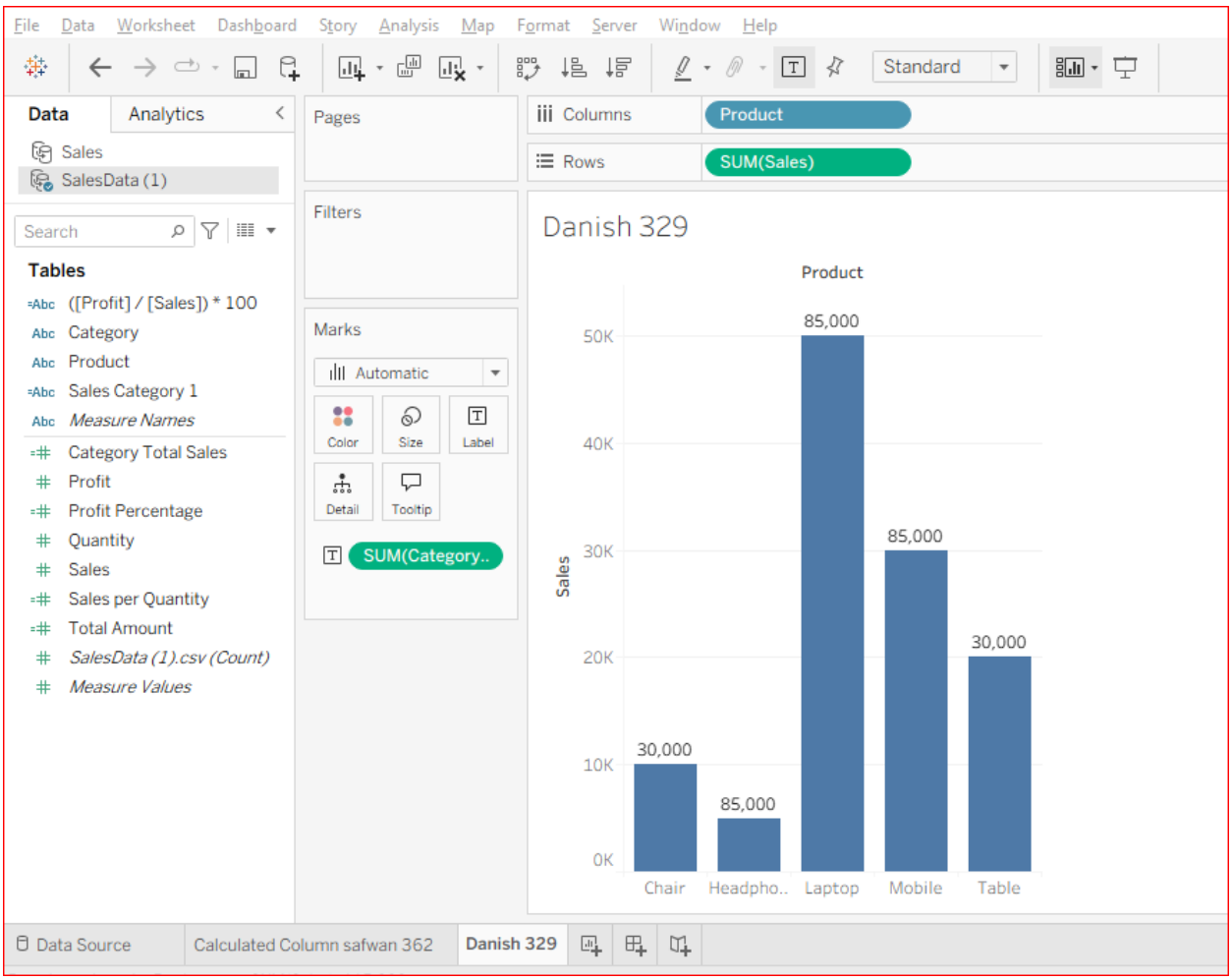
For **Electronics**:

50000 + 30000 = 80000

For **Furniture**:

10000 + 20000 = 30000

Notice that the **same category total appears for every product within that category**.

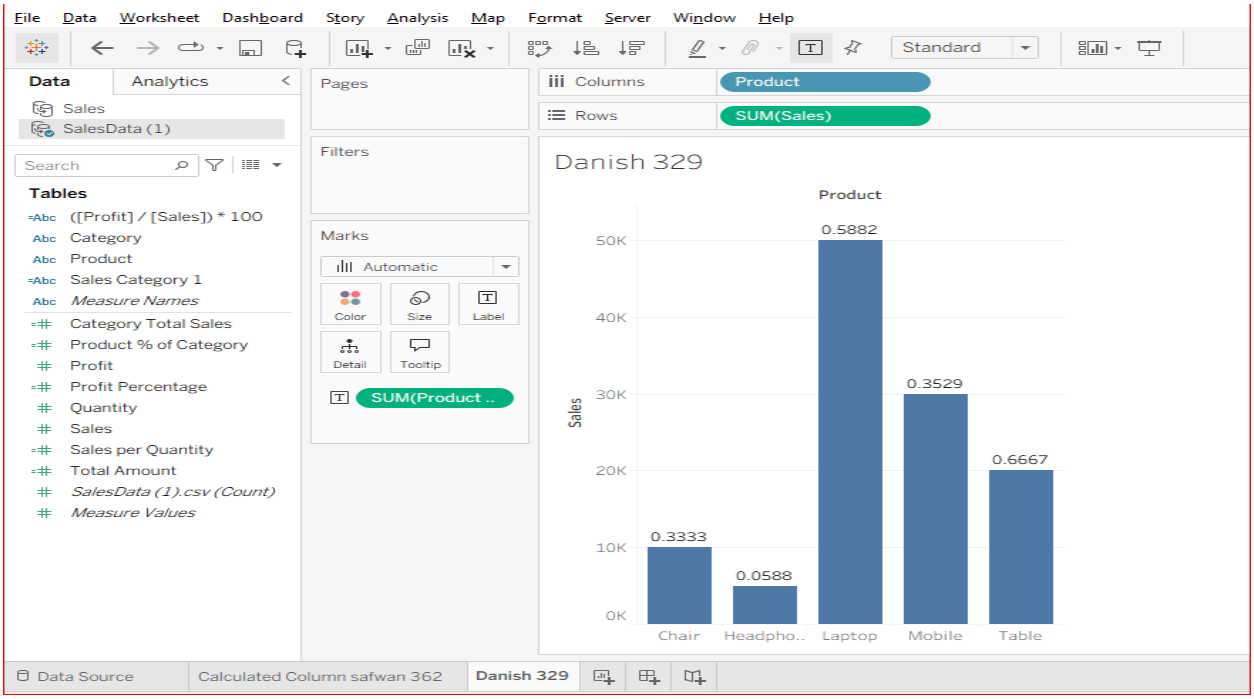


Calculate Product Percentage of Category Sales

Create another calculated field:

Product % of Category

[Sales] / { FIXED [Category] : SUM([Sales]) }



Use INCLUDE and EXCLUDE LOD expressions

INCLUDE LOD Expression

What is INCLUDE?

INCLUDE tells Tableau to **include an additional dimension** in the calculation, even if that dimension is not currently displayed in the view.

Syntax

{ INCLUDE [Dimension] : AGGREGATION([Measure]) }

Step 1: Create a Calculated Field

Go to:

Data Pane → Right-click → Create → Calculated Field

Name it:

Average Product Sales

Enter:

{ INCLUDE [Product] : AVG([Sales]) }

Click **OK**.

How it works

Suppose you create a view containing only **Category**.

Without INCLUDE, Tableau calculates the average at the Category level.

With:

{ INCLUDE [Product] : AVG([Sales]) }

Tableau first considers **Product**, then calculates the average.

For Electronics:

(50000 + 30000) / 2 = 40000

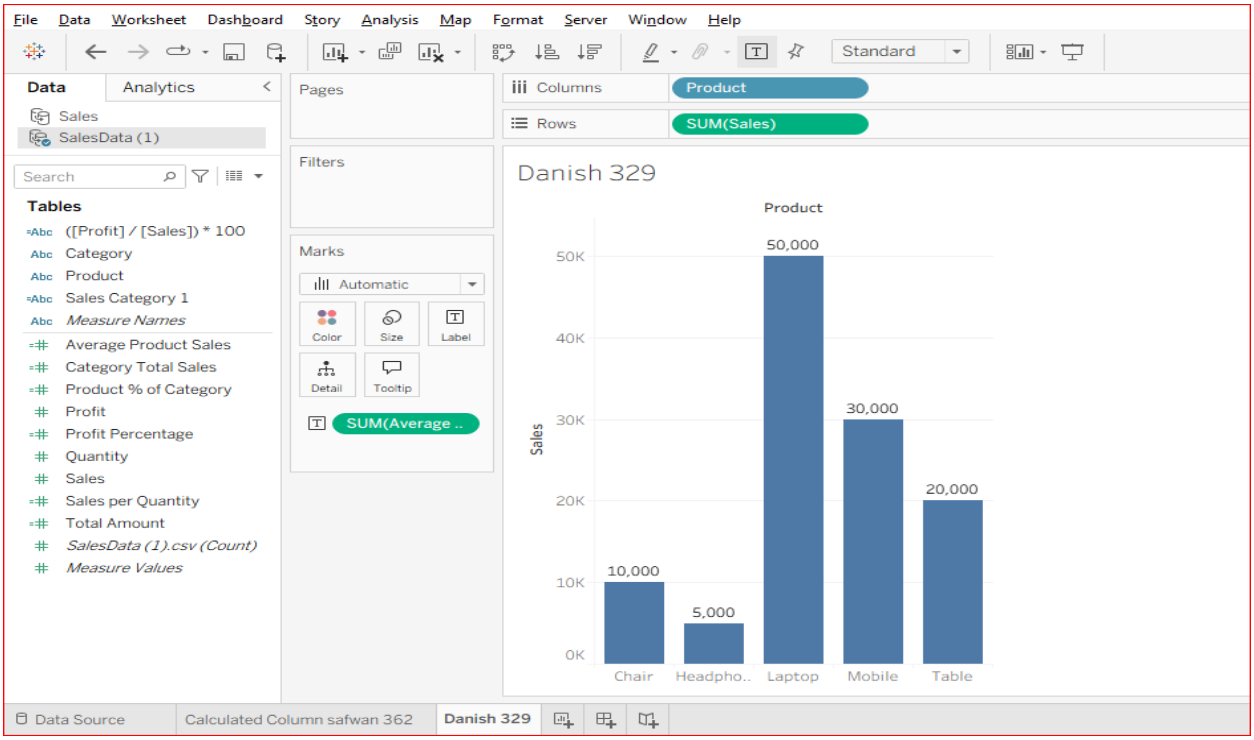
For Furniture:

(10000 + 20000) / 2 = 15000

In Tableau:

1. Open a new **Worksheet**.
2. Drag **Category → Rows**.
3. Drag your calculated field **Average Product Sales → Columns**.
4. Tableau will create a **bar chart**.

Output



EXCLUDE LOD Expression

What is EXCLUDE?

EXCLUDE tells Tableau to **remove a dimension from the calculation**, even if that dimension is present in the visualization.

Syntax

{ EXCLUDE [Dimension] : AGGREGATION([Measure]) }

Example

Create a calculated field named:

Category Total Sales

{ EXCLUDE [Product] : SUM([Sales]) }

Click **OK**.

INCLUDE vs EXCLUDE

Feature	INCLUDE	EXCLUDE
Purpose	Adds a dimension to calculation	Removes a dimension from calculation
Syntax	{ INCLUDE [Dimension] : SUM([Sales]) }	{ EXCLUDE [Dimension] : SUM([Sales]) }
Useful for	More detailed calculations	Higher-level calculations
Example	Include Product while viewing Category	Exclude Product while viewing Category

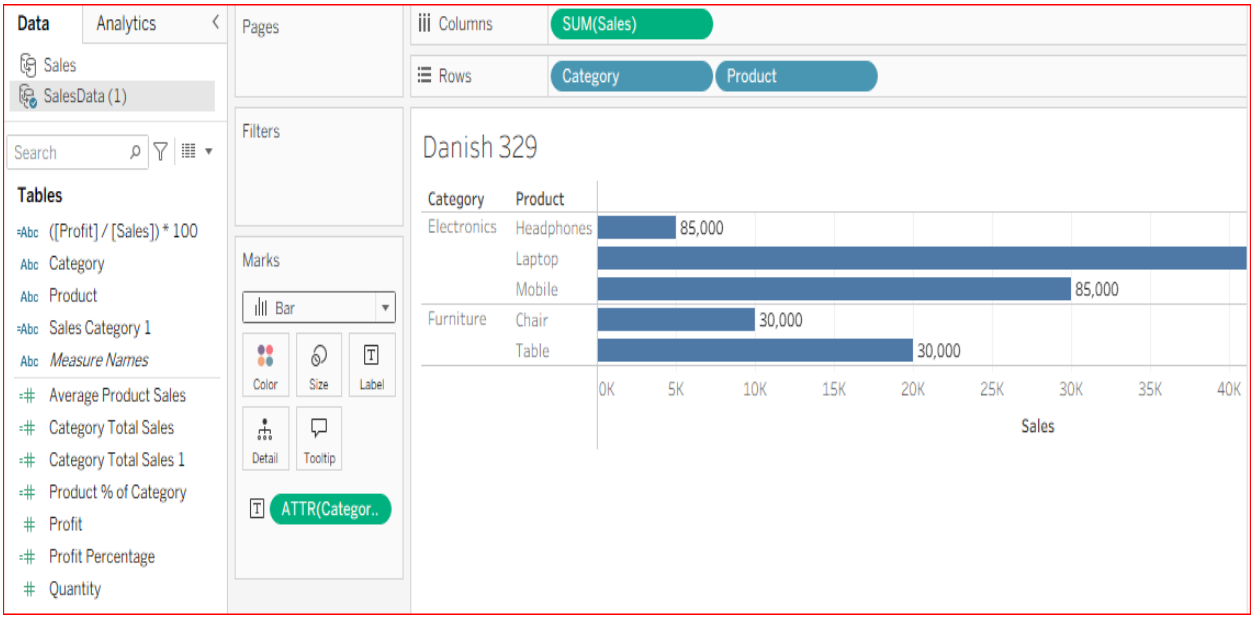
Visualize EXCLUDE LOD

Suppose you created:

{ EXCLUDE [Product] : SUM([Sales]) }

In Tableau:

1. Open another worksheet.
2. Drag **Category → Rows**.
3. Drag **Product → Rows** below Category.
4. Drag **Sales → Columns**.
5. Drag **Category Total Sales → Label** on the **Marks** card.
6. Select **Bar** from the Marks dropdown.



Create and Use Parameters for Dynamic Analysis in Tableau

Aim

To create and use **parameters** in Tableau to make the analysis and visualization dynamic.

What is a Parameter?

A **parameter** is a user-controlled value that allows you to change a calculation or visualization dynamically.

For example, we can create a parameter that allows the user to choose whether to display **Sales** or **Profit**.

Step 1: Create a Parameter

In Tableau:

1. Go to the **Data pane** on the left.
2. Click the small ▼ arrow near the Data pane.
3. Select **Create Parameter...**
4. Give it the name:

Select Measure

5. Set:

Data type: String

Allowable values: List

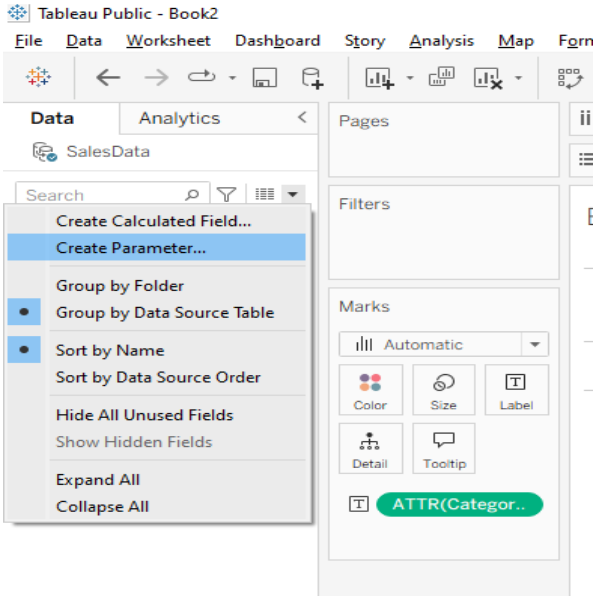
Add these values:

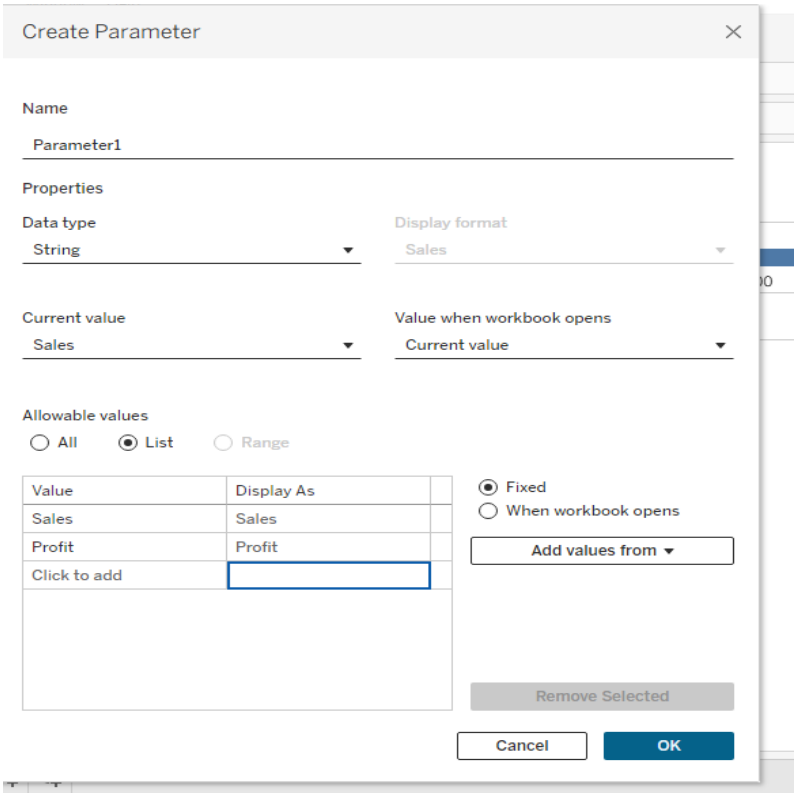
Sales

Profit

6. Click **OK**.

You will now see **Select Measure** under the **Parameters** section.





Create a Calculated Field

Right-click in the Data pane:

Create → Calculated Field

Name it:

Selected Measure

Enter:

```
CASE [Select Measure]
WHEN "Sales" THEN SUM([Sales])
WHEN "Profit" THEN SUM([Profit])
END
```

Click **OK**.

Step 3: Create the Visualization

Now:

1. Drag **Category → Rows**.
2. Drag **Selected Measure → Columns**.
3. Select **Bar Chart** from the Marks card or **Show Me**.

Your chart will initially show **Sales**.

Show the Parameter

Right-click **Select Measure** under Parameters.

Choose:

Show Parameter

A parameter control will appear on the worksheet.

You can now select:

Sales

or

Profit

When Sales is selected

The chart displays:

Category	Sales
Electronics	80,000
Furniture	30,000

When Profit is selected

The same chart automatically changes to:

Category	Profit
Electronics	14,000
Furniture	5,000

This is called **dynamic analysis** because the user can change the parameter and the visualization updates automatically.

